

# Parity Rock–Paper–Scissors Games and the Alper Constant

Alper Ferudun\*

5 October 2026

## Abstract

Two players choose numbers from  $\{1, \dots, n\}$ . Equal numbers draw, an odd difference is won by the smaller number, a nonzero even difference is won by the larger number, and a win with the number  $m$  pays  $2m - h$ , where  $0 \leq h < 2$  is fixed. For  $h = 1$  this is the game of Problem 1012 of Project Euler; for  $h = 0$  the payment is proportional to the winning number. For every  $h$  outside an explicit countable set we prove that the game has a unique equilibrium, supported on  $2k + 1$  consecutive numbers ending at  $n$  or  $n - 1$ , and we solve it in closed form. The probability  $P(n)$  of the largest number is  $((2k^2 + 1)R_k - 2(k^2 - 1))/(3(2k + 1))$  for an explicit product  $R_k$  that depends only on the largest payment, and the support widens at thresholds given by the zeros of these functions; for  $h \in \{0, 1\}$  the thresholds are cubic polynomials in  $k$ . For  $S_h(N) = \sum_{n=3}^N P(n)$  and  $h \in \{0, 1\}$  we prove  $S_h(N) = (3/2)^{4/3} N^{1/3} - \frac{1}{4} \log N - \mathcal{A}(h) - \frac{3}{2} \tau(1 - \tau) + O(N^{-1/3})$ , where  $\tau \in [0, 1]$  is the relative position of  $N$  in its block, and we determine the next term. The constant  $\mathcal{A}(h)$  plays the role that Euler’s constant plays for the harmonic series. We call  $\mathcal{A} = \mathcal{A}(0) = 1.7771799879\dots$  the Alper constant, give a series and an integral representation, and certify 80 decimal places of  $\mathcal{A}$  and of  $\mathcal{A}(1) = 1.6464176534\dots$  with interval arithmetic. No expression of either constant in classical constants was found.

## 1 Introduction

In rock–paper–scissors every option beats exactly one other option. With equal payments the unique equilibrium is uniform. Unequal payments change the picture in a counterintuitive way. Suppose the loser pays 5 to a winning rock, 3 to a winning paper and 1 to winning scissors. Then the unique equilibrium plays rock, paper and scissors with probabilities  $\frac{1}{9}$ ,  $\frac{5}{9}$  and  $\frac{1}{3}$ : the most lucrative option is used least.

We study the following family of extensions to  $n$  options. Two players choose numbers from  $\{1, \dots, n\}$  simultaneously. Equal numbers draw. If the numbers differ by an odd amount, the smaller number wins. If they differ by a nonzero even amount, the larger number wins. A win with the number  $m$  is paid  $2m - h$  by the loser, where  $0 \leq h < 2$  is a fixed parameter. We write  $G_n(h)$  for this game. Every increasing payment schedule  $c(m - \theta)$  with  $c > 0$  and  $0 \leq \theta < 1$  is covered after scaling, with  $h = 2\theta$ . Two members stand out:

- $h = 1$ : the payments are 1, 3, 5,  $\dots$ . For  $n = 3$  the numbers 1, 2, 3 play the roles of scissors, paper and rock with the payments 1, 3, 5 above. This game was posed as Problem 1012 of Project Euler [8], and a solution is described in [2].
- $h = 0$ : the payment is proportional to the winning number. This is the simplest member, and the one whose constant we call the Alper constant.

---

\*Mercury Software GmbH. Email: [alper@mercurycodelab.com](mailto:alper@mercurycodelab.com). GitHub: [AlperTheKing](https://github.com/AlperTheKing).

Let  $P(n)$  be the probability that the equilibrium of  $G_n(h)$  assigns to the largest number  $n$ , and put

$$S_h(N) = \sum_{n=3}^N P(n).$$

Solving  $G_n(h)$  numerically for moderate  $n$  suggests three patterns. The equilibrium always plays an odd number of consecutive numbers. Its width grows by two at isolated values of  $n$ . The ratio  $S_h(N)/N^{1/3}$  increases, but very slowly: for  $h = 0$  it is 1.330 at  $N = 10^3$  and 1.613 at  $N = 10^5$ , while its limit turns out to be  $(3/2)^{4/3} \approx 1.7171$ . The results below explain all three observations for the whole family. They also isolate the constant that governs the slow approach.

Throughout,  $\log$  is the natural logarithm,  $\gamma$  is Euler's constant and  $c_j = \binom{2j}{j} 4^{-j}$ . For  $k \geq 1$  and real  $y$  put

$$R_k(y) = \prod_{j=1}^k \frac{y - 2j}{y - 2j + 1}, \quad P_k(y) = \frac{(2k^2 + 1) R_k(y) - 2(k^2 - 1)}{3(2k + 1)}. \quad (1)$$

The variable  $y$  will be half of the largest payment in play, so  $y = n - \frac{h}{2}$  when the largest number is  $n$ . For  $k \geq 2$  the function  $P_k$  increases on  $y > 2k$  and has exactly one zero  $\beta_k$  there (Lemma 5.2). Put  $t_1(h) = 3$  and  $t_k(h) = \lfloor \beta_k + \frac{h}{2} \rfloor + 1$  for  $k \geq 2$ : this is the first integer  $n$  with  $P_k(n - \frac{h}{2}) > 0$ . The integers  $n$  with  $t_k(h) \leq n \leq t_{k+1}(h) - 1$  form the *block*  $k$ ; the blocks partition the integers  $n \geq 3$ . Finally let  $\mathcal{E}$  be the set of  $h \in [0, 2)$  for which  $\beta_k + \frac{h}{2}$  is an integer for some  $k \geq 2$ . It is countable, and it contains neither 0 nor 1.

**Theorem A** (Theorems 4.3 and 5.4). *Let  $0 \leq h < 2$ ,  $h \notin \mathcal{E}$ , and  $n \geq 3$ . The game  $G_n(h)$  has a unique equilibrium. Let  $k$  be the block of  $n$ . If  $n \leq t_{k+1}(h) - 2$ , the support is  $\{n - 2k, \dots, n\}$  and  $P(n) = P_k(n - \frac{h}{2}) > 0$ . If  $n = t_{k+1}(h) - 1$ , the support is  $\{n - 1 - 2k, \dots, n - 1\}$  and  $P(n) = 0$ . The whole equilibrium is given in closed form by Theorem 4.3. For the two main cases*

$$t_k(0) = \left\lceil \frac{4k^3 + 5k}{6} \right\rceil + 1, \quad t_k(1) = \left\lfloor \frac{4k^3 + 5k}{6} \right\rfloor + 2.$$

The first values are  $t_k(0) = 3, 8, 22, 47, 89, 150, 236, 349$  and  $t_k(1) = 3, 9, 22, 48, 89, 151, 236, 350$ .

**Theorem B** (Theorem 7.3). *Let  $h \in \{0, 1\}$  and  $\kappa = (3/2)^{4/3} = \sqrt[3]{81/16}$ . For  $N \geq 3$  let  $K$  be defined by  $t_K(h) - 1 \leq N < t_{K+1}(h) - 1$ , and put  $\tau = (N - t_K(h) + 1)/(t_{K+1}(h) - t_K(h)) \in [0, 1)$ . Then, uniformly in  $\tau$ ,*

$$S_h(N) = \kappa N^{1/3} - \frac{1}{4} \log N - \mathcal{A}(h) - \frac{3}{2} \tau(1 - \tau) - \frac{1 - 3\tau(1 - \tau)(4\tau - 1)}{4K} + O(N^{-2/3}).$$

Since  $K = (3N/2)^{1/3} + O(1)$ , the term with  $1/K$  is of order  $N^{-1/3}$ . The quadratic term cannot be dropped: within every block,  $S_h(N) - \kappa N^{1/3} + \frac{1}{4} \log N$  sweeps from about  $-\mathcal{A}(h)$  down to about  $-\mathcal{A}(h) - \frac{3}{8}$  and back. At the ends of the blocks, where  $P$  vanishes and  $\tau = 0$ , the oscillation disappears.

*The Alper constant.* The constant  $\mathcal{A}(h)$  is the exact analogue of Euler's constant. The harmonic numbers satisfy  $H_N - \log N \rightarrow \gamma$ ; here, along the ends  $N_K = t_{K+1}(h) - 1$  of the blocks,

$$\kappa N_K^{1/3} - \frac{1}{4} \log N_K - S_h(N_K) \longrightarrow \mathcal{A}(h) \quad (K \rightarrow \infty).$$

We define the *Alper constant* as the constant of the simplest member,  $h = 0$ :

$$\begin{aligned} \mathcal{A} = \mathcal{A}(0) = & 1.7771799879054231885280182162763415624255 \\ & 7103207026437429288159344562194194456353 \dots \end{aligned}$$

For the Project Euler game,

$$\begin{aligned} \mathcal{A}(1) = & 1.6464176534405600338544841159858132863029 \\ & 9683869991753007336025510008825020074857 \dots \end{aligned}$$

Section 8 expresses both constants as rapidly computable series and as integral-series, and explains how the 80 decimal places above are certified.

The case  $h = 1$  is a Project Euler problem, and its solvers may have obtained parts of our results for it, including the closed form and the first-order asymptotic formula; we make no priority claim for this case. To our knowledge, the family  $G_n(h)$  for  $h \neq 1$ , the transition law for general  $h$ , the rigorous proofs of the asymptotic formulas, the second-order term, the Alper constant and the certified digits are new. Section 2 sets up the games, Section 3 shows that the supports are intervals, and Section 4 solves the equilibrium equations on an interval of odd length. Section 5 proves the transition law and Theorem A. Section 6 sums the probabilities over blocks. Section 7 proves Theorem B, Section 8 studies the constants, and Section 9 compares the theorems with exact values of  $S_0(10^m)$  for  $m \leq 20$ .

## 2 The games

Let  $a_{ij}$  be the payment received by a player who chooses  $i$  when the opponent chooses  $j$ . It is 0 for  $i = j$ . Otherwise the winning number  $w$  is  $\min(i, j)$  when  $i - j$  is odd and  $\max(i, j)$  when  $i - j$  is even, and the winner receives  $2w - h$ .

**Lemma 2.1.** *For all  $i, j$ ,  $a_{ij} = (i - j) + \operatorname{sgn}(i - j)(-1)^{i+j}(i + j - h)$ . In particular  $a_{ji} = -a_{ij}$ .*

*Proof.* Take  $j < i$ . If  $i - j$  is even,  $i$  wins and the formula gives  $(i - j) + (i + j - h) = 2i - h$ . If  $i - j$  is odd,  $j$  wins and it gives  $(i - j) - (i + j - h) = -(2j - h)$ . The case  $j > i$  follows by antisymmetry.  $\square$

Against a vector  $p = (p_1, \dots, p_n)$  the pure option  $i$  earns  $f_i = \sum_j a_{ij}p_j$ , and we use this formula for every integer  $i$ . Since  $A = (a_{ij})$  is skew-symmetric,  $p^\top A p = 0$ , and the value of the game is 0. Hence a probability vector  $p$  is an equilibrium exactly when

$$p_i \geq 0, \quad \sum_i p_i = 1, \quad f_i \leq 0 \text{ for all } i, \quad f_i = 0 \text{ whenever } p_i > 0; \quad (2)$$

the last condition follows from the others because  $\sum_i p_i f_i = p^\top A p = 0$ . An equilibrium exists for every  $n$  by the minimax theorem [9]. The payoff rule does not depend on  $n$ , so the matrix of  $G_{n-1}(h)$  is a principal submatrix of the matrix of  $G_n(h)$ .

For  $n = 3$  the payoffs are  $f_1 = (2 - h)p_2 - (6 - h)p_3$ ,  $f_2 = -(2 - h)p_1 + (4 - h)p_3$  and  $f_3 = (6 - h)p_1 - (4 - h)p_2$ . Setting them to zero and normalising gives

$$p = \frac{(4 - h, 6 - h, 2 - h)}{12 - 3h},$$

so  $p = (\frac{1}{3}, \frac{1}{2}, \frac{1}{6})$  for  $h = 0$  and  $p = (\frac{1}{3}, \frac{5}{9}, \frac{1}{9})$  for  $h = 1$ , the weighted rock–paper–scissors equilibrium of the introduction.

## 3 Interval supports

For a probability vector  $p$  let  $X_i$  be the probability that option  $i$  wins: the total mass of the numbers  $j < i$  of the parity of  $i$  and of the numbers  $j > i$  of the other parity.

**Lemma 3.1.** *For every probability vector  $p$  and every  $i \leq n - 2$ ,*

$$f_{i+2} - f_i = 4(X_i - p_{i+1}) + (2i + 4 - h)(p_i + p_{i+2}) - (4i + 2 - 2h)p_{i+1}, \quad X_i \geq p_{i+1}.$$

*Proof.* The options  $i$  and  $i + 2$  have the same parity, so they obtain the same result against every  $j \notin \{i, i+1, i+2\}$ . When both win,  $i+2$  receives 4 more. The three remaining columns give  $a_{i+2,i} - a_{ii} = 2i + 4 - h$ ,  $a_{i+2,i+1} - a_{i,i+1} = -(2i + 2 - h) - (2i - h) = -(4i + 2 - 2h)$  and  $a_{i+2,i+2} - a_{i,i+2} = 2i + 4 - h$ . Option  $i$  beats  $i + 1$ , which gives  $X_i \geq p_{i+1}$ .  $\square$

**Proposition 3.2.** *Let  $n \geq 3$  and let  $p$  be an equilibrium of  $G_n(h)$ . Its support is an interval  $\{a, \dots, b\}$  with  $b \in \{n - 1, n\}$  and  $b > a$ .*

*Proof.* Suppose  $p_i > 0$  and  $p_{i+1} = 0$  for some  $i \leq n - 2$ . Then  $f_i = 0$  and Lemma 3.1 gives  $f_{i+2} \geq (2i + 4 - h)p_i > 0$ , which contradicts (2). So every chosen number below  $n - 1$  is followed by another chosen number, and the support is an interval  $\{a, \dots, b\}$  with  $b \geq n - 1 \geq 2$ . A single number  $b \geq 2$  is not an equilibrium, because  $b - 1$  beats it and receives  $2b - 2 - h > 0$ .  $\square$

The support has odd size; this is proved in Proposition 5.3. For the payments  $1, 3, 5, \dots$  of  $h = 1$  there is a short argument modulo 2, but for general  $h$  the parity of the support follows from the explicit solution of the next section.

## 4 The equilibrium on an interval of odd length

Fix  $k \geq 1$ ,  $b \geq 2k + 1$ ,  $a = b - 2k$ ,  $L = 2k + 1$ ,  $I = \{a, \dots, b\}$  and  $y = b - \frac{h}{2}$ , so that  $2y$  is the payment for a win with  $b$ . For a real vector  $p$  supported on  $I$  put

$$s = \sum_j p_j, \quad M = \sum_j j p_j, \quad q_i = \sum_{j \leq i} (-1)^{i-j} p_j \quad (i \in \mathbb{Z}), \quad Q = q_b.$$

Thus  $q_i = 0$  for  $i < a$ ,  $q_i = (-1)^{i-b}Q$  for  $i \geq b$ , and  $p_i = q_i + q_{i-1}$ . The numbers of the parity of  $b$  carry the mass  $\frac{s+Q}{2}$ , and the others carry  $\frac{s-Q}{2}$ .

**Lemma 4.1** (moment equation). *For  $a \leq j \leq b$ ,*

$$\sum_{i=a}^b a_{ij} = L \left( \frac{a+b}{2} - j \right) + \begin{cases} k, & j \equiv b \pmod{2}, \\ -(k+1), & j \not\equiv b \pmod{2}, \end{cases}$$

*independently of  $h$ . Consequently, if  $f_i = 0$  for all  $i \in I$ , then  $L(\frac{a+b}{2}s - M) + k\frac{s+Q}{2} - (k+1)\frac{s-Q}{2} = 0$ ; for  $s = 1$  this reads  $M - \frac{Q}{2} = \frac{a+b}{2} - \frac{1}{2L}$ , and for  $s = 0$  it reads  $M = \frac{Q}{2}$ .*

*Proof.* The column sum follows from Lemma 2.1 by pairing consecutive terms; the terms containing  $h$  cancel. Adding the  $L$  equations  $f_i = 0$  gives the stated equation.  $\square$

**Lemma 4.2** (neighbour identity). *For every real vector  $p$  supported on  $I$  and every integer  $i$ ,*

$$f_i + f_{i+1} = (2i + 1)s - 2M + (-1)^{b-i}Q + (2i + 2 - h)q_{i+1} - (2i - h)q_{i-1}.$$

*Proof.* With  $\varepsilon_j = (-1)^j$ , Lemma 2.1 gives  $f_i = is - M + \varepsilon_i[(i - h)T_i + U_i]$ , where  $T_i = \sum_{j < i} \varepsilon_j p_j - \sum_{j > i} \varepsilon_j p_j$  and  $U_i = \sum_{j < i} \varepsilon_j j p_j - \sum_{j > i} \varepsilon_j j p_j$ . Passing from  $i$  to  $i + 1$  changes  $T_i$  and  $U_i$  only in the terms of  $p_i$  and  $p_{i+1}$ , and  $\varepsilon_{i+1} = -\varepsilon_i$ , so  $f_i + f_{i+1} = (2i + 1)s - 2M - \varepsilon_i T_i - (2i + 1 - h)p_i + (2i + 2 - h)p_{i+1}$ . Finally  $\varepsilon_i T_i = q_i - q_{i-1} - (-1)^{b-i}Q$ , and  $p_i = q_i + q_{i-1}$ ,  $p_{i+1} = q_{i+1} + q_i$ .  $\square$

If  $s = 1$  and  $f_i = 0$  on  $I$ , the identity for  $a \leq i \leq b-1$  gives the  $2k$  neighbour equations

$$(2i+2-h)q_{i+1} - (2i-h)q_{i-1} = 2M - (2i+1) - (-1)^{b-i}Q \quad (a \leq i \leq b-1). \quad (3)$$

Each equation links  $q_{i-1}$  and  $q_{i+1}$ . The unknowns therefore split into the chain  $a-1, a+1, \dots, b-1$  and the chain  $a, a+2, \dots, b$ .

**Theorem 4.3.** *Let  $p$  be a real vector supported on  $I$  with  $\sum_j p_j = 1$  and  $f_i = 0$  for all  $i \in I$ . Put  $R = R_k(y)$ ,  $c_h = \frac{1-h}{3}$  and*

$$K = \frac{a+b-1}{2} - \frac{1}{2L}, \quad Q = \frac{2y(R-1) + 2k}{3} + \frac{1}{L}.$$

Then  $p_m = q_m + q_{m-1}$  with

$$q_m = K + c_h - \frac{m}{3} - \left(K + c_h - \frac{a-1}{3}\right) \prod_{\substack{a \leq i \leq m-1 \\ i \equiv b \pmod{2}}} \frac{2i-h}{2i+2-h} \quad (m = a-1, a+1, \dots, b-1), \quad (4)$$

$$q_m = K + c_h + Q - \frac{m}{3} + \left(\frac{b}{3} - K - c_h\right) \prod_{\substack{m+1 \leq i \leq b-1 \\ i \not\equiv b \pmod{2}}} \frac{2i+2-h}{2i-h} \quad (m = a, a+2, \dots, b). \quad (5)$$

In particular

$$p_b = P_k(y), \quad p_a = K + c_h + Q - \frac{a}{3} + \left(\frac{b}{3} - K - c_h\right) \frac{yR}{y-2k}.$$

*Proof.* On the first chain the equations (3) have  $(-1)^{b-i} = 1$ . The homogeneous step multiplies by  $\frac{2i-h}{2i+2-h}$ . A function  $\gamma' - \frac{m}{3}$  is a particular solution exactly when  $2\gamma' = 2M - 1 - Q + \frac{2-2h}{3}$ , because  $(2i+2-h)(i+1) - (2i-h)(i-1) = 6i+2-2h$ . This gives  $\gamma' = M - \frac{1+Q}{2} + c_h = K + c_h$ , by Lemma 4.1. Starting from  $q_{a-1} = 0$  gives (4). Over the whole chain the product is  $R_k(y)$ .

On the second chain  $(-1)^{b-i} = -1$ , the particular solution is  $K + c_h + Q - \frac{m}{3}$ , and the chain ends at  $q_b = Q$ . Running it downwards gives (5). The full product over the second chain is  $1/R_k(y+1)$ , and the telescoping identity

$$R_k(y) R_k(y+1) = \frac{y-2k}{y} \quad (6)$$

gives the formula for  $p_a = q_a$ .

It remains to determine  $Q$ . Add the equations (3) for  $a \leq i \leq b-1$ :

- the left sides telescope to  $(2b-h)Q + (2b-2-h)q_{b-1} - (2a+2-h)q_a - 2\sum_{m=a+1}^{b-2} q_m$ ;
- the terms  $\pm Q$  on the right cancel, and  $\sum_{i=a}^{b-1} (2i+1) = b^2 - a^2$ .

The normalisation  $1 = Q + 2\sum_{m=a}^{b-1} q_m$  removes the remaining sum. Together with Lemma 4.1 this yields

$$2y p_b - (2a-h)p_a = (2k-1)Q + \frac{1}{L}, \quad p_b = Q + q_{b-1}. \quad (7)$$

Inserting the two chain solutions leaves a linear equation for  $Q$  in which  $Q$  has the coefficient  $L$ . Its solution is the stated value. Then  $p_b = Q + q_{b-1} = P_k(y)$ .  $\square$

**Lemma 4.4** (the kernel). *For every  $0 \leq h < 2$  the kernel of the principal submatrix  $A_I$  is a line, spanned by a vector with nonzero coordinate sum. Consequently the formulas of Theorem 4.3 define a vector with  $\sum_j p_j = 1$  and  $f_i = 0$  on  $I$ , and it is the only such vector.*

*Proof.* A skew-symmetric matrix of odd order is singular, so  $A_I$  has a nonzero kernel vector  $w$ . Suppose that  $\sum_j w_j = 0$ . Lemma 4.1 gives  $M = \frac{Q}{2}$ , and Lemma 4.2 turns the equations  $f_i + f_{i+1} = 0$ ,  $a \leq i \leq b-1$ , into  $(2i+2-h)q_{i+1} - (2i-h)q_{i-1} = 2M - (-1)^{b-i}Q$ . On the first chain the right side is 0, so  $q \equiv 0$  there, because  $q_{a-1} = 0$ . On the second chain the right side is  $2Q$ ; the constant  $Q$  is a solution, and since  $q_b = Q$  the solution is this constant. Hence  $w_m = q_m + q_{m-1} = Q$  for all  $m \in I$ , so  $0 = \sum_j w_j = LQ$  and  $w = 0$ , a contradiction. Thus every nonzero kernel vector has nonzero sum and can be normalised; Theorem 4.3 shows that the normalised kernel vector is unique, so the kernel is a line.  $\square$

**Example 4.5.** Take  $h = 0$ ,  $b = 10$  and  $k = 2$ , so  $I = \{6, \dots, 10\}$ ,  $y = 10$  and  $R_2(10) = \frac{8}{9} \cdot \frac{6}{7} = \frac{16}{21}$ . The theorem gives

$$(p_6, p_7, p_8, p_9, p_{10}) = \left( \frac{29}{315}, \frac{92}{315}, \frac{34}{105}, \frac{74}{315}, \frac{2}{35} \right),$$

with  $Q = -\frac{17}{315}$  and  $M = \frac{496}{63}$ , so that  $M - \frac{Q}{2} = \frac{79}{10} = \frac{6+10}{2} - \frac{1}{10}$ . This is the equilibrium of  $G_{10}(0)$ , and  $P(10) = \frac{2}{35}$ .

## 5 Neighbouring options and the transition law

The interval solution decides its own fate through the two options next to the support.

**Proposition 5.1.** *In the setting of Theorem 4.3,*

$$f_{a-1} = \frac{2k+3}{2k+1} 2y R_k(y) P_{k+1}(y+1), \quad f_{b+1} = -\frac{2k+3}{2k+1} (2y+2) P_{k+1}(y+2).$$

*Proof.* Lemma 4.2 with  $i = a-1$  (where  $f_a = 0$ ,  $q_{a-2} = 0$  and  $(-1)^{b-a+1} = -1$ ) and with  $i = b$  (where  $f_b = 0$ ,  $q_{b+1} = -Q$  and  $q_{b-1} = p_b - Q$ ) gives

$$f_{a-1} = (2a-1) - 2M - Q + (2a-h)p_a, \quad f_{b+1} = (2b+1) - 2M - Q - 2y p_b.$$

By Lemma 4.1,  $2M + Q = a + b - \frac{1}{L} + 2Q$ , and by (7),

$$f_{a-1} = 2y p_b - L(1+Q), \quad f_{b+1} = L + \frac{1}{L} - 2Q - 2y p_b.$$

Inserting  $p_b = P_k(y)$  and the value of  $Q$  from Theorem 4.3, both payoffs become affine in  $R = R_k(y)$ :

$$\begin{aligned} 3L f_{a-1} &= (2k^2 + 4k + 3)(2y - 4k - 2) - 2k(k+2) 2y R, \\ 3L f_{b+1} &= 2k(k+2)(2y+2) - (2k^2 + 4k + 3) 2y R. \end{aligned}$$

On the other hand, the definition of  $R_{k+1}$  gives

$$R_{k+1}(y+1) R_k(y) = \frac{y-2k-1}{y}, \quad R_{k+1}(y+2) = \frac{y}{y+1} R_k(y).$$

Since  $2(k+1)^2 + 1 = 2k^2 + 4k + 3$  and  $2((k+1)^2 - 1) = 2k(k+2)$ , the definition (1) of  $P_{k+1}$  turns the right-hand sides into  $3(2k+3) 2y R P_{k+1}(y+1)$  and  $-3(2k+3)(2y+2) P_{k+1}(y+2)$ .  $\square$

**Lemma 5.2** (root bound). *Let  $k \geq 2$ . On the half-line  $y > 2k$  the function  $P_k$  is continuous and strictly increasing, and its unique zero  $\beta_k$  satisfies*

$$\frac{4k^3 + 5k + 3}{6} < \beta_k < \frac{4k^3 + 5k + 6}{6}.$$

*In particular  $\beta_k \notin \frac{1}{2}\mathbb{Z}$ . For  $k = 1$  we have  $P_1(y) = \frac{y-2}{3(y-1)} > 0$  for  $y > 2$ .*

*Proof.* For  $y > 2k$  every factor of  $R_k(y)$  is positive and increasing, so  $R_k$  increases strictly from 0 to 1, and  $P_k(y) = 0$  exactly when  $-\log R_k(y) = \lambda_k := \log \frac{2k^2+1}{2k^2-2}$ . Put  $x = y - k - \frac{1}{2}$  and

$$U = \bigcup_{j=1}^k [k - 2j + \frac{1}{2}, k - 2j + \frac{3}{2}], \quad G(x) = \int_U \frac{dt}{x+t}.$$

The interval with index  $j$  contributes  $\log \frac{y-2j+1}{y-2j}$ , so  $G(x) = -\log R_k(y)$ . The set  $U$  is symmetric about 0, has measure  $k$  and lies in  $[-d, d]$  with  $d = k - \frac{1}{2}$ ; on  $x > d$  the function  $G$  is strictly decreasing. Let  $x^*$  be the solution of  $G(x^*) = \lambda_k$ , so that  $\beta_k = x^* + k + \frac{1}{2}$ .

*Lower bound.* Let  $x_0 = k(4k^2 - 1)/6 > d$ . Then  $x_0 + \frac{k}{2} = \frac{k(2k^2+1)}{3}$  and  $x_0 - \frac{k}{2} = \frac{k(2k^2-2)}{3}$ , so  $\int_{-k/2}^{k/2} \frac{dt}{x_0+t} = \lambda_k$ . Folding both integrals onto  $t \geq 0$  gives

$$G(x_0) - \lambda_k = \int_{U_+} g(t) dt - \int_0^{k/2} g(t) dt, \quad g(t) = \frac{2x_0}{x_0^2 - t^2}, \quad U_+ = U \cap [0, \infty).$$

The set  $U_+$  has measure  $\frac{k}{2}$ , and since  $k \geq 2$  it has positive measure beyond  $\frac{k}{2}$ . As  $g$  is strictly increasing on  $[0, x_0)$ , the difference is positive. Hence  $G(x_0) > \lambda_k$  and  $x^* > x_0$ .

*Upper bound.* Let  $z = x_0 + \frac{1}{2}$ . By symmetry  $G(z) = \int_U \frac{z dt}{z^2 - t^2}$ , and for  $|t| \leq d$  we have  $\frac{z}{z^2 - t^2} \leq \frac{1}{z} (1 + \frac{t^2}{z^2 - d^2})$ . Since  $\int_U t^2 dt = kv$  with  $v = \frac{4k^2-3}{12}$ ,

$$G(z) \leq \frac{k}{z} \left( 1 + \frac{v}{z^2 - d^2} \right) < \frac{k}{x_0}.$$

The second inequality is equivalent to  $2x_0v < z^2 - d^2 = x_0^2 + x_0 - (k^2 - k)$ , which follows from  $x_0 > 2v$  and  $x_0 > k^2 - k$ . Finally  $\frac{k}{x_0} = \frac{6}{4k^2-1} = \frac{2u}{2+u}$  with  $u = \frac{3}{2k^2-2}$ , and  $\frac{2u}{2+u} < \log(1+u) = \lambda_k$ . Hence  $G(z) < \lambda_k$  and  $x^* < z$ .

Thus  $x_0 < x^* < x_0 + \frac{1}{2}$ , which is the stated bound because  $x_0 + k + \frac{1}{2} = \frac{4k^3+5k+3}{6}$ . The number  $\frac{4k^3+5k}{6}$  is an integer for even  $k$  and a half-integer for odd  $k$ , so the two bounds are consecutive points of  $\frac{1}{2}\mathbb{Z}$ .  $\square$

By Lemma 5.2,  $t_k(h) = \lfloor \beta_k + \frac{h}{2} \rfloor + 1$  is the first integer  $n$  with  $P_k(n - \frac{h}{2}) > 0$ , and for  $h \notin \mathcal{E}$  no integer  $n$  gives  $P_k(n - \frac{h}{2}) = 0$ . Since  $\beta_k \notin \frac{1}{2}\mathbb{Z}$ , neither 0 nor 1 lies in  $\mathcal{E}$ , and the bounds of Lemma 5.2 give the formulas for  $t_k(0)$  and  $t_k(1)$  stated in Theorem A, equivalently

$$t_k(h) = \frac{2}{3}k^3 + \frac{5}{6}k + \frac{5 + 2h - (-1)^{k+h}}{4} \quad (h \in \{0, 1\}). \quad (8)$$

**Proposition 5.3.** *Let  $h \notin \mathcal{E}$ . Every equilibrium of  $G_n(h)$  has a support of odd size  $2k + 1 \geq 3$ .*

*Proof.* Suppose the support  $I_0 = \{a, \dots, b\}$  of an equilibrium  $p$  has even size  $2k \geq 2$ , and let  $I = \{a, \dots, b+1\}$ ; the option  $b+1$  may lie outside the game, and  $f_{b+1}$  is computed with Lemma 2.1. On  $I$  we have  $A_I p = \varphi e_{b+1}$  with  $\varphi = f_{b+1}$ , because  $f_i = 0$  on  $I_0$ . Let  $v$  be the normalised kernel vector of  $A_I$  from Lemma 4.4. By Theorem 4.3,  $v_{b+1} = P_k(b+1 - \frac{h}{2})$ , which is nonzero: for  $k \geq 2$  because  $h \notin \mathcal{E}$ , and for  $k = 1$  by Lemma 5.2. Skew-symmetry gives  $0 = -(A_I v)^\top p = v^\top A_I p = \varphi v_{b+1}$ , so  $\varphi = 0$ . Then  $p$  is a normalised kernel vector of  $A_I$ , so  $p = v$  and  $p_{b+1} = v_{b+1} \neq 0$ , a contradiction.  $\square$

**Theorem 5.4.** *Let  $h \notin \mathcal{E}$  and let  $n \geq 3$  lie in the block  $k$ . The game  $G_n(h)$  has exactly one equilibrium. If  $n \leq t_{k+1}(h) - 2$ , its support is  $\{n - 2k, \dots, n\}$  and  $P(n) = P_k(n - \frac{h}{2}) > 0$ . If  $n = t_{k+1}(h) - 1$ , its support is  $\{n - 1 - 2k, \dots, n - 1\}$  and  $P(n) = 0$ .*

*Proof.* Write  $t_k$  for  $t_k(h)$ . Let  $p$  be an equilibrium of  $G_n(h)$ ; it exists by the minimax theorem. By Propositions 3.2 and 5.3 its support is  $\{a, \dots, b\}$  with  $b \in \{n-1, n\}$  and  $b-a = 2k'$  for some  $k' \geq 1$ , and by Theorem 4.3,  $p_b = P_{k'}(b - \frac{h}{2})$ .

First,  $t_{k'} \leq b \leq t_{k'+1} - 2$ . Indeed  $P_{k'}(b - \frac{h}{2}) = p_b > 0$ , so  $b \geq t_{k'}$ . If  $a \geq 2$ , the option  $a-1$  exists and  $f_{a-1} \leq 0$ . As  $R_{k'}(b - \frac{h}{2}) > 0$ , Proposition 5.1 gives  $P_{k'+1}(b+1 - \frac{h}{2}) \leq 0$ , hence  $< 0$ , and so  $b+1 < t_{k'+1}$ . If  $a = 1$ , then  $b = 2k' + 1 \leq t_{k'+1} - 2$ , because  $t_{k'+1} > \beta_{k'+1} > 2k' + 3$  by Lemma 5.2.

If  $b = n-1$ , the option  $n = b+1$  exists and  $f_n \leq 0$ , so  $P_{k'+1}(b+2 - \frac{h}{2}) \geq 0$  by Proposition 5.1, hence  $> 0$ , and  $b+2 \geq t_{k'+1}$ . Together with the first step,  $n = t_{k'+1} - 1$ ; this is the last element of the block  $k'$ , so  $k' = k$ ,  $n = t_{k+1} - 1$  and  $P(n) = p_n = 0$ .

If  $b = n$ , the first step gives  $t_{k'} \leq n \leq t_{k'+1} - 2$ , so  $k' = k$ ,  $n \leq t_{k+1} - 2$  and  $P(n) = p_n = P_k(n - \frac{h}{2}) > 0$ .

The two cases are distinguished by whether  $n = t_{k+1} - 1$ , so the support is determined by  $n$ . By Lemma 4.4,  $p$  is the only normalised vector on this support with vanishing payoffs on it. Hence the equilibrium is unique.  $\square$

*Remark 5.5.* The proof never checks that the closed-form probabilities are positive or that the excluded options lose. Positivity and the sign conditions are inherited from the equilibrium whose existence the minimax theorem guarantees; as a by-product, every  $p_m$  in Theorem 4.3 is positive when  $t_k(h) \leq b \leq t_{k+1}(h) - 2$ .

Write  $\beta_k = \frac{4k^3+5k+3}{6} + \vartheta_k$  with  $0 < \vartheta_k < \frac{1}{2}$ . Unwinding the definition of  $t_k(h)$ ,

$$\mathcal{E} = \{1 - 2\vartheta_k : k \geq 2 \text{ even}\} \cup \{2 - 2\vartheta_k : k \geq 3 \text{ odd}\}.$$

For even  $k$ ,  $t_k(h)$  equals  $t_k(0)$  when  $h < 1 - 2\vartheta_k$  and  $t_k(1)$  when  $h > 1 - 2\vartheta_k$ . For odd  $k$ ,  $t_k(0) = t_k(1)$ , and  $t_k(h)$  equals this common value when  $h < 2 - 2\vartheta_k$  and exceeds it by one when  $h > 2 - 2\vartheta_k$ . Numerically  $\vartheta_k$  decreases like  $\frac{3}{8k}$  ( $\vartheta_2 = 0.1458$ ,  $\vartheta_{10} = 0.0372$ ,  $\vartheta_{400} = 0.00094$ ), so  $\mathcal{E}$  accumulates only at 1 and 2, and for every fixed  $h \notin \mathcal{E}$  the thresholds eventually agree with those of  $h = 0$  (if  $h < 1$ ) or of  $h = 1$  (if  $h \geq 1$ ). For  $h \in \mathcal{E}$  uniqueness fails: if  $\beta_k + \frac{h}{2} = m \in \mathbb{Z}$ , letting  $h$  tend to this value from both sides in Theorem 5.4 shows that the interval solutions with supports  $\{m-2k+1, \dots, m-1\}$  and  $\{m-2k, \dots, m-2\}$  are both equilibria of  $G_{m-1}(h)$ , since limits of equilibria are equilibria.

## 6 Block sums

From now on  $h \in \{0, 1\}$ . By Theorem 5.4,  $S_h$  is a sum over blocks. For  $k \geq 1$  put

$$\ell_k = t_{k+1}(h) - t_k(h) - 1 = 2k^2 + 2k + \frac{1 + (-1)^{k+h}}{2}, \quad B_k = B_k(h) = \sum_{n=t_k(h)}^{t_{k+1}(h)-2} P_k\left(n - \frac{h}{2}\right),$$

so that  $S_h(t_{K+1}(h) - 1) = \sum_{k=1}^K B_k$ . Table 1 lists the first blocks for  $h = 0$ .

**Lemma 6.1** (partial fractions). *For  $k \geq 1$ ,*

$$1 - R_k(y) = \sum_{j=1}^k \frac{w_{k,j}/2}{y - 2j + 1}, \quad w_{k,j} = 2(2k - 2j + 1) c_{j-1} c_{k-j}, \quad \sum_{j=1}^k w_{k,j} = 2k.$$

Moreover  $w_{k,j}/(2k) = \Pr(X = j-1)$ , where  $X$  has the beta-binomial distribution with parameters  $k-1$ ,  $\frac{1}{2}$  and  $\frac{3}{2}$ , so that

$$P_k(y) = \frac{1}{2k+1} - \frac{2k(2k^2+1)}{3(2k+1)} \mathbb{E}\left[\frac{1}{2y-2-4X}\right]. \quad (9)$$

| $k$ | $t_k(0)$ | $n$ with $P(n) = P_k(n) > 0$ | zero of $P$ | $\ell_k$ | $B_k(0)$      |
|-----|----------|------------------------------|-------------|----------|---------------|
| 1   | 3        | 3–6                          | 7           | 4        | 0.90555555556 |
| 2   | 8        | 8–20                         | 21          | 13       | 1.2051685518  |
| 3   | 22       | 22–45                        | 46          | 24       | 1.2449235895  |
| 4   | 47       | 47–87                        | 88          | 41       | 1.3328846236  |
| 5   | 89       | 89–148                       | 149         | 60       | 1.3441658999  |
| 6   | 150      | 150–234                      | 235         | 85       | 1.3842642085  |

Table 1: The first blocks of the game  $G_n(0)$ . In the block  $k$  the support has  $2k + 1$  elements; at the zero of  $P$  it is shifted down by one.

*Proof.*  $R_k$  is a rational function of  $y$  with simple poles at  $y = 2j - 1$  and  $R_k(\infty) = 1$ . The factor with index  $j$  equals  $1 - \frac{1}{y-2j+1}$ . At its pole the other factors take the value  $\prod_{i < j} \frac{2(j-i)-1}{2(j-i)} \prod_{i > j} \frac{2(i-j)+1}{2(i-j)} = c_{j-1}(2k - 2j + 1)c_{k-j}$ , which gives the residues. Since  $\sum_q (2q + 1)c_q z^q = (1 - z)^{-3/2}$  and  $\sum_q c_q z^q = (1 - z)^{-1/2}$ , comparing coefficients in  $(1 - z)^{-3/2}(1 - z)^{-1/2} = (1 - z)^{-2}$  gives  $\sum_{q=0}^{k-1} (2q + 1)c_q c_{k-1-q} = k$ . The beta-binomial probability of  $x$  is  $\binom{k-1}{x} (\frac{1}{2})_x (\frac{3}{2})_{k-1-x} / (2)_{k-1}$  in Pochhammer notation [7]; with  $(\frac{1}{2})_x = x! c_x$  and  $(\frac{3}{2})_z = (2z + 1) z! c_z$  this equals  $c_x(2k - 2x - 1)c_{k-1-x}/k$ . Finally  $2(y - 2j + 1) = 2y - 2 - 4(j - 1)$  and  $P_k = \frac{1}{2k+1} - \frac{2k^2+1}{3(2k+1)}(1 - R_k)$ .  $\square$

For example  $w_{2,\cdot} = (3, 1)$ ,  $w_{3,\cdot} = (\frac{15}{4}, \frac{3}{2}, \frac{3}{4})$  and  $w_{4,\cdot} = (\frac{35}{8}, \frac{15}{8}, \frac{9}{8}, \frac{5}{8})$ . Let  $\psi$  be the digamma function,  $H_m = \sum_{i \leq m} \frac{1}{i} = \psi(m+1) - \psi(1)$  the harmonic numbers and  $H_m^{\text{odd}} = \sum_{i \leq m} \frac{1}{2i-1} = \frac{1}{2}(\psi(m + \frac{1}{2}) - \psi(\frac{1}{2}))$  the odd harmonic numbers [1, 5.4.13, 5.4.15].

**Proposition 6.2.** For  $k \geq 1$ , with  $t_k = t_k(h)$ ,

$$B_k = \frac{\ell_k}{2k+1} - \frac{2k^2+1}{3(2k+1)} \sum_{j=1}^k \frac{w_{k,j}}{2} \left[ \psi\left(t_{k+1} - \frac{h}{2} - 2j\right) - \psi\left(t_k - \frac{h}{2} - 2j + 1\right) \right].$$

The bracket equals  $H_{t_{k+1}-2j-1} - H_{t_k-2j}$  for  $h = 0$  and  $2(H_{t_{k+1}-2j-1}^{\text{odd}} - H_{t_k-2j}^{\text{odd}})$  for  $h = 1$ . In particular  $B_k$  is rational,  $B_1(0) = \frac{163}{180}$ , and  $S_0(10) = B_1(0) + P_2(8) + P_2(9) + P_2(10) = \frac{25493}{25200}$ .

*Proof.* Sum Lemma 6.1 over the block, using  $\sum_{n=A}^B \frac{1}{n-\frac{h}{2}-2j+1} = \psi(B-\frac{h}{2}-2j+2) - \psi(A-\frac{h}{2}-2j+1)$ .  $\square$

**Proposition 6.3** (integral representation). For  $k \geq 1$  let

$$W_k(x) = \sum_{j=1}^k w_{k,j} x^{4(k-j)} = \frac{4k}{\pi} \int_0^1 \theta^{1/2} (1 - \theta)^{-1/2} (1 - \theta + \theta x^4)^{k-1} d\theta.$$

Then

$$B_k = \frac{\ell_k}{2k+1} - \frac{2k^2+1}{3(2k+1)} \int_0^1 x^{2t_k-h-4k+1} \frac{1-x^{2\ell_k}}{1-x^2} W_k(x) dx.$$

*Proof.* Insert  $\frac{1}{y-2j+1} = \int_0^1 x^{2n-h-4j+1} dx$  (with  $y = n - \frac{h}{2}$ ) into the block sum and sum the geometric series over the block. The integral formula for  $W_k$  is  $2k$  times the probability generating function of  $k - 1 - X \sim \text{BetaBinomial}(k - 1, \frac{3}{2}, \frac{1}{2})$  at  $x^4$ , written as a binomial mixture over a  $\text{Beta}(\frac{3}{2}, \frac{1}{2})$  parameter, whose normalising constant is  $B(\frac{3}{2}, \frac{1}{2}) = \frac{\pi}{2}$ .  $\square$

*Remark 6.4.* For fixed  $k$  the summand  $P_k(n - \frac{h}{2})$  has no rational antidifference in  $n$ : at infinity it equals  $\frac{1}{2k+1} - \frac{k(2k^2+1)}{3(2k+1)n} + O(n^{-2})$ , while  $G(n+1) - G(n)$  has no term in  $1/n$  for any rational function  $G$ . This is why the digamma function enters Proposition 6.2; in particular the ordinary rational Gosper certificate [5] does not exist.

## 7 Asymptotics

Let  $h \in \{0, 1\}$ , fix  $k$ , and write  $\eta = (-1)^{k+h}$ ,

$$D = \frac{4}{3}k^3, \quad a_\eta = \frac{5}{3}k + \frac{1-\eta}{2}, \quad \text{so that} \quad 2t_k(h) - h - 2 = D + a_\eta$$

by (8), and  $\ell_k = 2k^2 + 2k + \frac{1+\eta}{2}$ . Everything below depends on  $h$  and the parity of  $k$  only through  $\eta$ . For  $n = t_k(h) + r$  with  $0 \leq r < \ell_k$ , equation (9) with  $y = n - \frac{h}{2}$  becomes

$$P_k\left(n - \frac{h}{2}\right) = \frac{1}{2k+1} \left[1 - \left(1 + \frac{1}{2k^2}\right) \mathbb{E}\left[\frac{1}{1 + \xi_r/D}\right]\right], \quad \xi_r = a_\eta + 2r - 4X, \quad (10)$$

because  $\frac{2k(2k^2+1)}{3D} = 1 + \frac{1}{2k^2}$ . Put  $T_m = \sum_{r=0}^{\ell_k-1} \mathbb{E}[\xi_r^m]$ . The beta-binomial law has  $\mathbb{E}X = \frac{k-1}{4}$  and  $\text{Var}(4X) = k^2 - 1$  [7], so

$$T_0 = \ell_k, \quad T_1 = \ell_k \mu, \quad T_2 = \ell_k \left[ \mu^2 + \frac{\ell_k^2 - 1}{3} + k^2 - 1 \right], \quad \mu = 2k^2 + \frac{8}{3}k + 1.$$

In general  $T_m$  is, for each  $\eta$ , a polynomial in  $k$  of degree at most  $2m + 2$  with rational coefficients. It follows from the factorial moments  $\mathbb{E}[X(X-1)\cdots(X-i+1)] = (k-1)(k-2)\cdots(k-i) \frac{(\frac{1}{2})_i}{(2)_i}$  and the power sums  $\sum_{r < \ell_k} r^j$ .

**Lemma 7.1.** *For  $k \geq 8$ ,  $0 \leq r < \ell_k$  and  $0 \leq X \leq k-1$  we have  $|\xi_r/D| \leq 4/k$ .*

*Proof.* We have  $-\frac{7}{3}k + 4 \leq \xi_r \leq 4k^2 + \frac{17}{3}k$ , and both  $(4k^2 + \frac{17}{3}k)/D = \frac{3}{k} + \frac{17}{4k^2}$  and  $\frac{7}{3}k/D = \frac{7}{4k^2}$  are at most  $\frac{4}{k}$ .  $\square$

**Proposition 7.2.** *Let  $\eta = (-1)^{k+h}$ .*

(a)  $B_k = \frac{3}{2} - \frac{3}{4k} + \frac{3\eta}{8k^2} + O(k^{-3})$ .

(b) *For every  $J \geq 0$  there are rational numbers  $b_{j,\pm}$ , independent of  $h$ , with  $B_k = \sum_{j=0}^J b_{j,\eta} k^{-j} + O_J(k^{-J-1})$ . Table 2 lists them for  $j \leq 8$ .*

(c) *For  $k \geq 8$  and  $M \geq 1$ ,*

$$\left| B_k - \frac{1}{2k+1} \left[ \ell_k - \left(1 + \frac{1}{2k^2}\right) \sum_{m=0}^M (-1)^m \frac{T_m}{D^m} \right] \right| \leq \frac{3}{2} k \frac{(4/k)^{M+1}}{1 - 4/k}.$$

*Proof.* Sum (10) over  $r$  and use  $\frac{1}{1+u} = \sum_{m=0}^M (-u)^m + \frac{(-u)^{M+1}}{1+u}$ . By Lemma 7.1 the remainder is at most  $\frac{\ell_k}{2k+1} (1 + \frac{1}{2k^2}) \frac{(4/k)^{M+1}}{1-4/k}$ , and  $\ell_k \leq 3k^2$  gives (c). For each  $\eta$ ,  $T_m/D^m = O(k^{2-m})$  is a Laurent polynomial in  $k$ , so the bracket in (c) equals  $k F_{M,\eta}(1/k)$  for a polynomial  $F_{M,\eta}$  with rational coefficients, and  $B_k = F_{M,\eta}(1/k)/(2 + 1/k) + O(k^{-M})$ . The term with index  $m$  contributes only from the power  $k^{1-m}$  on, so the Taylor coefficients of  $F_{M,\eta}(z)/(2+z)$  up to  $z^{M-1}$  do not depend on  $M$ . This gives (b).

For (a) take  $M = 3$ . The formulas for  $T_1$  and  $T_2$  and  $T_3 = \sum_r \mathbb{E}[\xi_r^3] = 32k^8 + O(k^7)$  give

$$\frac{\ell_k}{2k^2} = 1 + \frac{1}{k} + O(k^{-2}), \quad \frac{T_1}{D} = 3k + 7 + \frac{25 + 3\eta}{4k} + O(k^{-2}), \quad \frac{T_2}{D^2} = 6 + \frac{21}{k} + O(k^{-2}), \quad \frac{T_3}{D^3} = \frac{27}{2k} + O(k^{-2}).$$

The bracket in (c) equals  $-\frac{\ell_k}{2k^2} + (1 + \frac{1}{2k^2}) \left( \frac{T_1}{D} - \frac{T_2}{D^2} + \frac{T_3}{D^3} \right) = 3k - \frac{3(1-\eta)}{4k} + O(k^{-2})$ . Dividing by  $2k+1$  gives (a).  $\square$

| $j$ | $b_{j,+}$    | $b_{j,-}$    |
|-----|--------------|--------------|
| 0   | 3/2          | 3/2          |
| 1   | -3/4         | -3/4         |
| 2   | 3/8          | -3/8         |
| 3   | -9/20        | 57/40        |
| 4   | 291/160      | -219/160     |
| 5   | -2859/560    | -327/280     |
| 6   | 6417/560     | 17793/2240   |
| 7   | -203421/8960 | -168561/8960 |
| 8   | 390009/8960  | 274299/8960  |

Table 2: Coefficients of the expansion  $B_k \sim \sum_j b_{j,\eta} k^{-j}$  of Proposition 7.2, with  $\eta = (-1)^{k+h}$ : the column  $b_{j,+}$  serves even  $k$  for  $h = 0$  and odd  $k$  for  $h = 1$ . The certificates store the coefficients for  $j \leq 59$ .

Put  $e_k = e_k(h) = B_k - \frac{3}{2} + \frac{3}{4k}$ . By Proposition 7.2(a),  $e_k = (-1)^{k+h} \frac{3}{8k^2} + O(k^{-3})$ , so  $\sum_k e_k$  converges absolutely, and  $\sum_{k>K} e_k = O(K^{-2})$ .

**Theorem 7.3.** *Let  $h \in \{0, 1\}$ ,  $N \geq 3$ , and let  $K$  and  $\tau$  be as in Theorem B. Then*

$$S_h(N) = \kappa N^{1/3} - \frac{1}{4} \log N - \mathcal{A}(h) - \frac{3}{2} \tau(1 - \tau) - \frac{1 - 3\tau(1 - \tau)(4\tau - 1)}{4K} + O(K^{-2}), \quad (11)$$

uniformly in  $\tau$ , where

$$\mathcal{A}(h) = \frac{3}{2} + \frac{3}{4}\gamma + \frac{1}{4} \log \frac{3}{2} - \sum_{k=1}^{\infty} \left( B_k(h) - \frac{3}{2} + \frac{3}{4k} \right). \quad (12)$$

At the block ends  $N_K = t_{K+1}(h) - 1$  this reads  $S_h(N_K) = \kappa N_K^{1/3} - \frac{1}{4} \log N_K - \mathcal{A}(h) - \frac{1}{4K} + O(K^{-2})$ .

*Proof. Block ends.* By Theorem 5.4,  $S_h(N_K) = \sum_{k \leq K} B_k = \frac{3}{2}K - \frac{3}{4}H_K + \sum_{k \leq K} e_k$ , where  $H_K = \log K + \gamma + \frac{1}{2K} + O(K^{-2})$ . Hence

$$S_h(N_K) = \frac{3}{2}K - \frac{3}{4} \log K - \frac{3}{4}\gamma - \frac{3}{8K} + \sum_{k \geq 1} e_k + O(K^{-2}).$$

By (8),  $N_K = \frac{2}{3}(K+1)^3 + \frac{5}{6}(K+1) + O(1)$ , so  $\kappa N_K^{1/3} = \frac{3}{2}(K+1) + \frac{5}{8K} + O(K^{-2})$  and  $\log N_K = \log \frac{2}{3} + 3 \log K + \frac{3}{K} + O(K^{-2})$ . Combining these expansions with (12) gives the block-end formula, because  $-\frac{3}{8} - \frac{5}{8} + \frac{3}{4} = -\frac{1}{4}$ .

*General  $N$ .* The case  $K = 1$  is a finite statement, so let  $K \geq 2$ . Put  $m = t_K(h) - 1 = N_{K-1}$ ,  $r = N - m$  and  $D_K = t_{K+1}(h) - t_K(h) = 2K^2 + 2K + O(1)$ , so that  $\tau = r/D_K$ . The block-end formula for  $K - 1$  gives  $S_h(m) = \kappa m^{1/3} - \frac{1}{4} \log m - \mathcal{A}(h) - \frac{1}{4K} + O(K^{-2})$ . For  $1 \leq s \leq r$ , equation (10) with  $\xi = a_\eta + 2s - 2 - 4X$  applies to  $n = t_K(h) - 1 + s$ . Here  $\mathbb{E}\xi = \frac{2}{3}K + 2s - \frac{1+\eta}{2}$  and  $\mathbb{E}\xi^2 = (\mathbb{E}\xi)^2 + K^2 - 1$ , and expanding  $\frac{1}{1+u}$  to second order with Lemma 7.1 gives, uniformly in  $1 \leq s < D_K$ ,

$$P_K\left(t_K(h) - 1 + s - \frac{h}{2}\right) = \frac{3s}{4K^4} - \frac{3s}{8K^5} - \frac{9s^2}{8K^7} + O(K^{-4}).$$

Summing over  $s \leq r$  and inserting  $r = \tau D_K$ ,

$$S_h(N) - S_h(m) = \frac{3}{2}\tau^2 + \frac{\frac{9}{4}\tau^2 - 3\tau^3}{K} + O(K^{-2}).$$

Since  $r/m = O(1/K)$ , Taylor expansion gives

$$\kappa N^{1/3} - \kappa m^{1/3} = \frac{3}{2}\tau + \frac{3(\tau - \tau^2)}{2K} + O(K^{-2}), \quad \frac{1}{4} \log \frac{N}{m} = \frac{3\tau}{4K} + O(K^{-2}).$$

Adding up, the coefficient of  $1/K$  is  $-\frac{1}{4} - \frac{3}{4}\tau + \frac{15}{4}\tau^2 - 3\tau^3 = -\frac{1}{4}(1 - 3\tau(1 - \tau)(4\tau - 1))$ , and the terms of order one give  $-\frac{3}{2}\tau(1 - \tau)$ .  $\square$

Theorem B follows, because  $K \asymp N^{1/3}$ .

## 8 The Alper constant

**Definition 8.1.** For  $h \in \{0, 1\}$  let  $N_K = t_{K+1}(h) - 1$  be the zeros of  $P$ , that is, the ends of the blocks. Put

$$\mathcal{A}(h) = \lim_{K \rightarrow \infty} \left( \kappa N_K^{1/3} - \frac{1}{4} \log N_K - S_h(N_K) \right).$$

The *Alper constant* is  $\mathcal{A} = \mathcal{A}(0)$ .

By Theorem 7.3 the limit exists and is given by the series (12). In the same way as Euler's constant measures how far the harmonic numbers lie from  $\log N$ , the Alper constant measures how far the equilibrium sums  $S_0(N)$  lie from their smooth approximation  $\kappa N^{1/3} - \frac{1}{4} \log N$ ; along general  $N$  the parabola  $\frac{3}{2}\tau(1 - \tau)$  of Theorem B has to be added.

### Representations

The series (12) has rational terms  $e_k = B_k - \frac{3}{2} + \frac{3}{4k}$ , which Proposition 6.2 evaluates exactly, and it converges like  $\sum k^{-2}$  with alternating signs. Its leading part sums in closed form: since  $\sum_k (-1)^{k+1} k^{-2} = \frac{\pi^2}{12}$ , we have  $\sum_k (-1)^{k+h} \frac{3}{8k^2} = (-1)^{h+1} \frac{\pi^2}{32}$ . Thus

$$\sum_{k=1}^{\infty} e_k(h) = (-1)^{h+1} \frac{\pi^2}{32} + \sum_{k=1}^{\infty} \left( e_k(h) - (-1)^{k+h} \frac{3}{8k^2} \right),$$

and the last series, with terms  $O(k^{-3})$ , equals

$$\begin{aligned} &0.5655231753318100092599469738976772679775991822265765812903645817609304 \dots \quad (h = 0), \\ &0.0794352347285882502563253866959460981430671626443740113590515840927113 \dots \quad (h = 1). \end{aligned}$$

Subtracting further terms of Proposition 7.2 produces the contributions  $\frac{381}{320}\zeta(3)$  and  $-\frac{1497}{1280}\zeta(4)$  for  $h = 0$ , and  $-\frac{69}{320}\zeta(3)$  and  $\frac{2073}{1280}\zeta(4)$  for  $h = 1$ . Each such step is an exact identity and accelerates convergence, but we do not claim that the expansion in  $1/k$  converges. Its coefficients in Table 2 grow quickly. Finally, Proposition 6.3 turns (12) into a series of explicit integrals.

### Certified digits

**Theorem 8.2.** *The 80 decimal places of  $\mathcal{A} = \mathcal{A}(0)$  and of  $\mathcal{A}(1)$  printed in the introduction are correct. Moreover*

$$\begin{aligned} \sum_{k \geq 1} e_k(0) &= 0.2570980377977675524213691301515475449990 \\ &\quad 4607575030187421494741375405402095091354 \dots \end{aligned}$$

*with all printed digits correct.*

*Method.* For each  $h$ , split  $\sum_k e_k$  at  $K_0 = 192$ . The 191 head terms come from Proposition 6.2. In the tail,  $B_k$  is replaced by the expansion of Proposition 7.2 through  $j = 59$ , with the exact rational coefficients computed from the polynomials  $T_m$ . The tail sums are Hurwitz zeta values: if  $a_0$  and  $a_1$  are the least even and odd integers  $\geq K_0$ , then  $\sum_{k \geq K_0, k \equiv a_i (2)} k^{-j} = 2^{-j} \zeta(j, a_i/2)$ . Two errors remain. Proposition 7.2(c) with  $M = 60$  bounds the geometric truncation in each block. The Taylor truncation leaves  $G(z)/(2+z)$ , where  $G = F_{60,\eta} - (2+z) \sum_{j \leq 59} b_{j,\eta} z^j$  is an explicit polynomial divisible by  $z^{60}$ , so it is at most  $\frac{1}{2} \sum_p |g_p| k^{-p}$  in terms of the coefficients  $g_p$  of  $G$ . Both errors are summed over  $k \geq K_0$  with  $\sum_{k \geq K_0} k^{-s} \leq K_0^{-s} + K_0^{1-s}/(s-1)$ . The two bounds are below  $3.481 \cdot 10^{-100}$  and  $1.26 \cdot 10^{-109}$  for both values of  $h$ .

All transcendental quantities ( $\psi$ , Hurwitz  $\zeta$ ,  $\gamma$  and  $\log \frac{3}{2}$ ) are evaluated in Arb ball arithmetic [6] at 160 digits, via python-flint [4]. The resulting enclosures have radius below  $7.4 \cdot 10^{-100}$ , and the program checks that the rounding to 80 places is constant on each ball. For  $h = 0$  a second run with  $K_0 = 128$ , the expansion through  $j = 47$  and 110 digits gives an enclosure of radius below  $6 \cdot 10^{-71}$  that overlaps the first. For both  $h$ , an independent 130-digit mpmath summation differs from the certified midpoint by less than  $10^{-93}$ . This is a computer-assisted evaluation of the error bounds above, not a formal proof.  $\square$

## Is there a closed form?

We searched for an integer relation between  $\mathcal{A}$  and

$$1, \gamma, \log 2, \log 3, \pi, \pi^2, \pi^4, \zeta(3), \zeta(5), G, \\ \log \pi, \log \Gamma(\tfrac{1}{4}), \log \Gamma(\tfrac{1}{3}), \sqrt{2}, \pi\sqrt{3}, \log^2 2, \log^2 3, \log 2 \log 3, \pi \log 2,$$

where  $G$  is Catalan's constant, using PSLQ [3] in 120-digit arithmetic with tolerance  $10^{-85}$  and coefficients of absolute value at most 1000. The same search was run for  $\mathcal{A}(1)$ , for  $\mathcal{A}(h) - \frac{1}{4} \log \frac{3}{2}$  and for  $\sum_k e_k(h)$ , and also against the subset  $1, \gamma, \log 2, \log 3, \pi^2, \zeta(3)$ . No relation was found. This is evidence, not a proof, that  $\mathcal{A}$  has no simple closed form.

Two facts make  $\mathcal{A}$  less arbitrary than an empirical constant. First, the block ends  $t_k(0)$  are exact polynomials in  $k$  for each parity, by (8); the fractional positions of the zeros  $\beta_k$  never enter. Second, every ingredient of (12) is explicit: rational block sums and the constants  $\gamma$  and  $\log \frac{3}{2}$ .

## How the constants were found

The constant  $\mathcal{A}(1)$  was seen first, numerically. Exact block sums give  $S_1(10^m)$  to many digits for  $m \leq 20$ . After  $\kappa N^{1/3}$  is removed, the remainder still drifts like  $-\frac{1}{4} \log N$ . After this drift is also removed, it oscillates inside each block, and the parabola  $\frac{3}{2} \tau(1-\tau)$  accounts for the oscillation; what is left clusters near 1.6464. The proofs did not use the special payments  $2m-1$  except in the explicit form of the thresholds, which led to the family  $G_n(h)$  and to its simplest member  $h = 0$ , whose constant is the Alper constant. Theorem 7.3 and the certified evaluations turned these observations into theorems.

## 9 Numerical data

Table 3 lists  $S_0(10^m)$  for  $1 \leq m \leq 20$ , computed in two ways. The certified computation uses Arb balls: for  $m \leq 9$  it adds all block sums of Proposition 6.2, and for  $m \geq 10$  it uses  $S(N) = \frac{3}{2}(K-1) - \frac{3}{4}H_{K-1} + \sum_k e_k - \sum_{k \geq K} e_k$  plus the partial block, with the tail and the partial block taken from Proposition 7.2. It certifies 30 decimals. The second computation adds all block sums in 100-digit decimal arithmetic, switching to a bounded moment expansion of each block for  $k \geq 1000$ . The two

agree to within  $5 \cdot 10^{-31}$ . The fourth column shows the normalised residual  $K \delta(N)$  of the first-order formula, where

$$\delta(N) = S_0(N) - \left( \kappa N^{1/3} - \frac{1}{4} \log N - \mathcal{A} - \frac{3}{2} \tau(1 - \tau) \right),$$

and the last column shows its limit  $c(\tau) = -\frac{1}{4}(1 - 3\tau(1 - \tau)(4\tau - 1))$  predicted by Theorem 7.3. The difference between the two columns is  $O(1/K)$ , and it is below  $10^{-5}$  for  $m \geq 15$ .

| $N$       | $S_0(N)$                     | $S_0(N)/N^{1/3}$ | $K \delta(N)$ | $c(\tau)$ |
|-----------|------------------------------|------------------|---------------|-----------|
| $10^1$    | 1.01162698412698412698       | 0.469555651331   | -0.164628     | -0.268039 |
| $10^2$    | 4.76382820302819364808       | 1.026335673796   | -0.203027     | -0.275258 |
| $10^3$    | 13.30208678375677871897      | 1.330208678376   | -0.089062     | -0.151146 |
| $10^4$    | 32.56774386666910234077      | 1.511660762675   | 0.041852      | 0.017932  |
| $10^5$    | 74.87679399375861732781      | 1.613171624585   | -0.287017     | -0.291100 |
| $10^6$    | 166.10210752872543471053     | 1.661021075287   | -0.085058     | -0.089064 |
| $10^7$    | 363.77124561032651935744     | 1.688476551614   | 0.013286      | 0.010878  |
| $10^8$    | 790.28053769454961122398     | 1.702607805266   | -0.197141     | -0.198360 |
| $10^9$    | 1709.80689936028358017628    | 1.709806899360   | 0.034865      | 0.034247  |
| $10^{10}$ | 3691.53391596882046613243    | 1.713458260326   | -0.269046     | -0.269115 |
| $10^{11}$ | 7961.51938003709654146938    | 1.715257353771   | -0.223351     | -0.223461 |
| $10^{12}$ | 17161.84551749443685308992   | 1.716184551749   | -0.289401     | -0.289424 |
| $10^{13}$ | 36983.76129919255133262475   | 1.716634134713   | -0.291166     | -0.291168 |
| $10^{14}$ | 79689.45681287532869805299   | 1.716857301875   | -0.114766     | -0.114791 |
| $10^{15}$ | 171696.35811503857501272212  | 1.716963581150   | -0.122306     | -0.122312 |
| $10^{16}$ | 369920.57702331276152428602  | 1.717019219635   | -0.270994     | -0.270996 |
| $10^{17}$ | 796982.05832200729109138505  | 1.717045793782   | -0.228871     | -0.228872 |
| $10^{18}$ | 1717058.94943545492359063880 | 1.717058949435   | -0.254104     | -0.254105 |
| $10^{19}$ | 3699305.29382848385618983787 | 1.717065414396   | -0.286261     | -0.286261 |
| $10^{20}$ | 7969925.78324778285942282425 | 1.717068458441   | -0.016978     | -0.016979 |

Table 3:  $S_0(N)$  rounded to 20 decimals, the ratio  $S_0(N)/N^{1/3}$ , and the second-order residual of Theorem 7.3. The limit of the ratio is  $\kappa = 1.7170713\dots$

## 10 Concluding remarks

The structure of the equilibria rests on two features of the payments: they grow linearly, which makes the payoff differences in Lemma 3.1 positive and the neighbour equations solvable, and the largest payment enters only through  $y = n - \frac{h}{2}$ . For  $h \notin \mathcal{E} \cup \{0, 1\}$ , Remark 5.5 indicates that the thresholds eventually coincide with those of  $h = 0$  or  $h = 1$ ; the proof of Theorem 7.3 should then give the same first-order formula, with a constant  $\mathcal{A}(h)$  defined by the actual block sums, while the term of order  $1/K$  acquires an  $h$ -dependent part. We have not pursued this.

We list some questions that we have not answered.

1. Does  $\mathcal{A}$  have a closed form in classical constants? Is it irrational?
2. How does  $\mathcal{A}(h)$  depend on  $h$ ? Its definition changes at the points of  $\mathcal{E}$ , which accumulate at 1 and 2.
3. What is the nature of the series  $\sum_j b_{j,\pm} k^{-j}$  in Table 2? Its coefficients grow quickly, and we expect it to diverge.
4. The method of Theorem 7.3 should give further terms of  $S_h(N)$  in powers of  $1/K$ , with polynomial coefficients in  $\tau$ . A general description of these polynomials would be interesting.

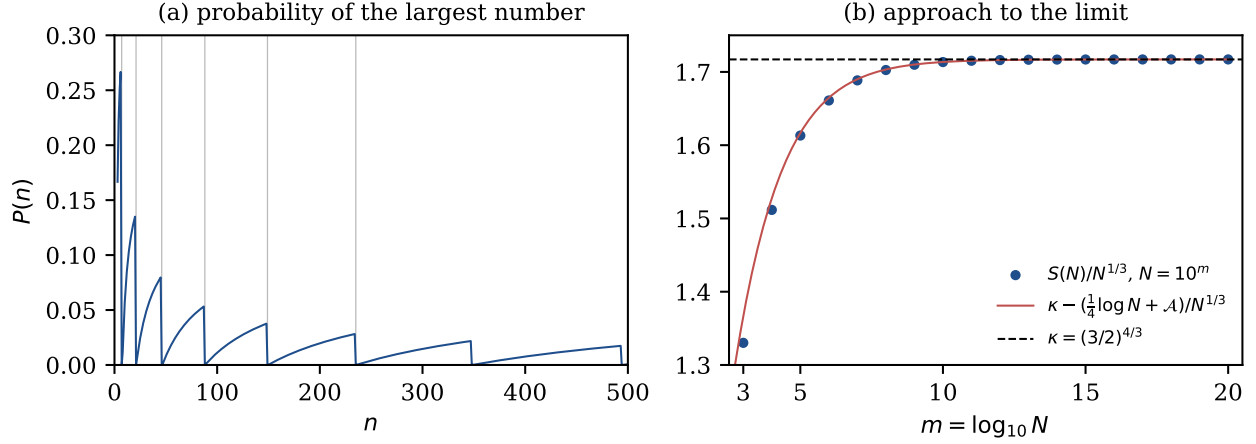


Figure 1: The game  $G_n(0)$ . (a) The probability  $P(n)$  of the largest number for  $3 \leq n \leq 500$ ; the grey lines mark the zeros  $n = t_k(0) - 1$ . Within each block  $P$  increases, and it drops to 0 when the support widens. (b)  $S_0(N)/N^{1/3}$  at  $N = 10^m$  for  $3 \leq m \leq 20$  (dots), the limit  $\kappa$  (dashed), and the curve  $\kappa - (\frac{1}{4} \log N + \mathcal{A})/N^{1/3}$ .

The exact checks of all displayed identities and examples, the computation of the tables and the figure, and the programs that certify  $\mathcal{A}$  and  $\mathcal{A}(1)$  are included in the source archive and at <https://github.com/AlperTheKing/alper-constant>. AI assistance was used in the derivations, in writing and running the verification code, and in preparing this manuscript. The results have not been refereed or verified in a proof assistant. This is an unrefereed preprint.

## References

- [1] NIST Digital Library of Mathematical Functions. F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl and M. A. McClain, eds. <https://dlmf.nist.gov/>.
- [2] EulerSolve. *Problem 1012: Rock Paper Scissors*, solution notes. <https://eulersolve.org/problem/1012/> (accessed 5 October 2026).
- [3] H. R. P. Ferguson, D. H. Bailey and S. Arno. Analysis of PSLQ, an integer relation finding algorithm. *Math. Comp.* **68** (1999), 351–369. doi:10.1090/S0025-5718-99-00995-3.
- [4] The FLINT team. *FLINT: Fast Library for Number Theory*, and its Python interface *python-flint*. <https://flintlib.org>.
- [5] R. W. Gosper, Jr. Decision procedure for indefinite hypergeometric summation. *Proc. Natl. Acad. Sci. USA* **75** (1978), 40–42. doi:10.1073/pnas.75.1.40.
- [6] F. Johansson. Arb: efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Trans. Comput.* **66** (2017), 1281–1292. doi:10.1109/TC.2017.2690633.
- [7] N. L. Johnson, A. W. Kemp and S. Kotz. *Univariate Discrete Distributions*, 3rd ed. Wiley, Hoboken, 2005.

- [8] Project Euler. *Problem 1012: Rock Paper Scissors*. <https://projecteuler.net/problem=1012> (accessed 5 October 2026).
- [9] J. von Neumann. Zur Theorie der Gesellschaftsspiele. *Math. Ann.* **100** (1928), 295–320.  
[doi:10.1007/BF01448847](https://doi.org/10.1007/BF01448847).